

Guidance and surveillance capability for bystander *Staphylococcus aureus* tetracycline resistance under doxy-PEP: a three-stream evidence audit

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Synopsis

Background. Doxycycline post-exposure prophylaxis (doxy-PEP) prevents bacterial sexually transmitted infections and has scaled into guidance internationally. Doxycycline also selects tetracycline resistance beyond its target, and *Staphylococcus aureus*—a commensal and pathogen for which doxycycline is one of a limited number of oral options—is the natural bystander.

Objectives. To determine whether the guidance and surveillance deployed around doxy-PEP could follow the bystander-resistance signal the pivotal trial's final analysis resolved, rather than to estimate a population effect.

Methods. A reproducible, public-data audit of three evidence streams—clinical trials, governmental guidance and deployed US surveillance—each scored against one yardstick: the within-exposed relative risk it could detect, benchmarked to an empirical colonization estimate.

Results. The final randomized DoxyPEP analysis reported a 3.89-fold higher hazard of incident doxycycline-resistant *S. aureus* under doxy-PEP (95% CI 1.42–10.68; 68 of 393 versus 5 of 163 incident events), with no effect on colonization clearance. Yet 0 of 6 outbreak-matched

governmental guidelines (0 of 10 coded) required *S. aureus* monitoring, and 0 of 12 established US surveillance systems linked doxy-PEP exposure to an *S. aureus* tetracycline phenotype at a common population denominator. Under realistic exposure density and isolate volumes, an ordinary population system would need an implausible within-exposed effect ($RR \approx 6\text{--}28$) to move a rate.

Conclusions. The upstream trial generated information the downstream architecture is not built to receive. Following the signal requires sentinel surveillance linking doxy-PEP exposure to the *S. aureus* tetracycline phenotype at a common denominator. No population-level causal claim is made.

Introduction

Doxycycline post-exposure prophylaxis (doxy-PEP) substantially reduces bacterial sexually transmitted infections (STIs) and has moved quickly from trial result into guidance at every level, from the World Health Organization through national and municipal health departments.^{1,2} Every dose is selective pressure, and not only on the organisms it targets. *Staphylococcus aureus* is the consequential bystander: it colonizes the anatomical sites these trials swabbed and carries mobile tetracycline resistance determinants.³ For two decades, community MRSA in these networks has been documented not as steady-state carriage but as clustered, outbreak-pattern transmission—the USA300 lineage moving through sexual networks in San Francisco, Tokyo and Chicago^{4,5}; the organism colonizing these networks is the same one carried on skin community-wide, so resistance selected within the network could, if transmitted, extend beyond it. Tetracycline-resistant USA300 was in fact documented in an MSM-predominant community health population two decades ago, although by the doxycycline-sparing *tet(K)* mechanism, with those isolates remaining doxycycline-susceptible.⁶ Doxycycline is one of a limited number of oral options for outpatient skin and soft-tissue infection, a choice often made empirically before any methicillin result returns,^{7,8} so tetracycline susceptibility can matter clinically before methicillin status is known.

Clinicians who prescribe doxy-PEP have registered the concern directly: in a survey of Ital-

ian STI-clinic dermatologists, 91.7% named methicillin-resistant *S. aureus* (MRSA) selection as a worry, tied with gonococcal resistance as the highest-rated,⁹ and a systematic review of 1840 healthcare professionals across four countries likewise placed antimicrobial resistance among the leading concerns about doxy-PEP.¹⁰ The trial literature has now answered a version of it. In the final DoxyPEP analysis, among participants free of doxycycline-resistant *S. aureus* at baseline, doxy-PEP carried a 3.89-fold higher hazard of incident doxycycline-resistant *S. aureus* (95% CI 1.42–10.68; 68/393 versus 5/163; $P = 0.0044$), while the hazard of clearing colonization did not differ between arms (HR 1.01, 0.69–1.46).¹¹ The trial’s authors called for public-health surveillance of antimicrobial resistance (AMR) in *S. aureus* and other commensals.

That call defines the present question. We ask not whether doxy-PEP drives population-level resistance—a claim we do not make—but whether the guidance and surveillance deployed around doxy-PEP could follow the signal the trial resolved, at the population level where the clinical stake lies. The clinically decisive axis is tetracycline (doxycycline) susceptibility used in empirical oral therapy, not the methicillin axis around which staphylococcal surveillance is organized. We audit the three places an answer would have to come from—the trials, the guidance and the deployed surveillance—each against a single detectability yardstick.

Materials and methods

Design

The study is a documentary and quasi-computational audit assembled in three streams, collected and analyzed in sequence: clinical trials (Stream A), the governmental guidance built on them (Stream B) and the population surveillance meant to monitor them (Stream C). No individual-level or restricted data were used; every input is public, so no ethical approval was required. Sources were located, read, coded with page/section/table locators and SHA-256-pinned before analysis; analysis decisions that could go either way were date-logged before results were known; and every figure and table regenerates from a single command over a test-gated pipeline archived with a citable DOI.¹² A large language model assisted with writing and

analysis engineering; the author verified every numerical value and quotation against primary
sources and takes full responsibility (see Acknowledgements).

Detectability yardstick

Each stream is scored on one commensurable quantity: the within-exposed relative risk (RR)
it could detect. The empirical benchmark is Soge *et al.*¹³ (tetracycline-resistant *S. aureus* colo-
nization, 18% versus 8% in doxy-PEP users with >3 doses/month versus non-users; $RR \approx 2.25$);
a cross-organism gonococcal $RR \approx 1.42$ is carried as a flagged, more conservative secondary.
Because the randomized trial reports a within-exposed hazard ratio of 3.89, well above this
band, the detectability verdict below is invariant to which anchor is used: the relative risk
a population system would need to see (RR 6–28, Results) exceeds both the cross-sectional
benchmark and the trial’s own hazard ratio.

Sources and coding

The author identified, retrieved and screened all source documents for eligibility; each eligible
document was then coded from full text against a prespecified schema, every value carrying a
page, section or table locator. For the guidance corpus (Stream B), two independent machine
(large language model) passes coded each document under separate blinded prompts, and dis-
agreements were resolved by the author against the source. Per-field agreement is reported as
raw concordance, which measures the reproducibility of the coding protocol rather than inde-
pendent human adjudication; a Cohen’s κ is given for context only, with the caveat that the
two passes share a model lineage and are therefore not fully independent (Limitations). The
surveillance enumeration (Stream C) was coded by a single pass; its inclusion and exclusion
list, with a stated reason per system, is given in the Supplementary data and reflects systems as
deployed as of August 2026. All coding prompts and outputs are in the public repository.

Stream A—trials

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For each primary-trial *S. aureus* arm comparison (first-party published data only), we computed, from arm sizes and observed control rate, the exact (Fisher) power to detect the externally fixed benchmark effect and the minimum detectable RR at 80% power; fixing the effect externally makes this a design-based sensitivity analysis rather than observed power. The *S. aureus* assay is doxycycline throughout (E-test, MIC $\geq$ 16 mg/L). We tabulated how the resistance result was reported across trial venues and secondary syntheses (numerator, denominator basis, conditioning set and terminology). The final trial performed the appropriate participant-level time-to-event test itself; we cite that randomized result directly and did not re-estimate it. Cross-sectional carriage cannot resolve the clustered transmission structure of *S. aureus* in these networks; the supporting dispersion, trend-test degradation and outbreak-shape analyses are MRSA-scoped (MRSA being a subset, not the analytic unit) and are reported in the Supplementary data (available as Supplementary data at JAC Online).

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Stream B—guidance

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To make the denominator principled, we anchored it to the same clustering evidence used in Stream A: US jurisdictions with documented community-associated MRSA/USA300 transmission in men who have sex with men.^{4,5} We took governmental public-health guidance only (clinic and provider-organization documents excluded, with reasons). Each document was coded from full text, one record per document, every finding carrying a locator, for whether it requires, suggests or omits *S. aureus* monitoring and whether it names, counsels on or measures the organism (coding provenance above). An independence check separated genuinely independent declines from documents that defer to, or are adapted from, one another.

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Stream C—surveillance

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We enumerated a closed universe of established US governmental surveillance systems (federal, state or metropolitan) that could plausibly capture, at a population denominator, either doxy-PEP/doxycycline exposure or *S. aureus* AMR phenotype, and coded each on three ca-

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pabilities: exposure-side capture, phenotype-side capture (of the *S. aureus* tetracycline unit, not the MRSA subset) and the join, recording the geographic resolution of each side. A supporting analysis asked whether, even at a common denominator, the exposed subgroup is dense enough to move a population rate. The exposed population is indexed on the combined PrEP and people-living-with-HIV male count (primary specification; the trial enrolled both, AIDSVu 2024 state PrEP and prevalence), so the dilution fraction is $f = \phi((\text{male-PrEP} + \text{male-PLWH}) \text{ density} / 10^5)$ uptake $d \kappa$, where d is the share of users taking >3 doses/month (the subgroup carrying the benchmark effect) and κ the isolate-enrichment factor; f is population-independent because it tracks exposure density, not headcount, and the minimum detectable change in a proportion near baseline R_0 inverts to $\text{RR}_{\text{needed}} = 1 + \text{MDE} / (f R_0)$. Parameters were not uniformly generous: isolate volume was set generously high (favoring detection), while the realistic under-sampling ($\kappa = 0.5$) and dose-threshold ($d = 0.5$) factors bound against it. Both ranges and the resulting $\text{RR}_{\text{needed}}$ are reported; full grids and the metropolitan breakdown even are in the Supplementary data.

Results

Stream A: trial measurement and reporting of the *S. aureus* endpoint

When the analysis matched the question, the signal appeared. The final DoxyPEP analysis ran the participant-level test the interim had not—time to first detection of doxycycline-resistant *S. aureus* by arm, standard-of-care censored at crossover, by Cox proportional hazards—and returned HR 3.89 (95% CI 1.42–10.68; $P = 0.0044$), with colonization clearance unchanged (HR 1.01).¹¹ The 2023 interim, by contrast, was presented as reassuring on a cross-sectional resistance fraction and was underpowered for any externally specified effect (0 of 4 comparisons powered; minimum detectable RR 3.8–5.1 against the benchmark 1.42–2.25). The hazard ratio is not comparable to that cross-sectional minimum: the estimand, not the sample size alone, became adequate to the question.

Two measurement features of the interim record persist and matter downstream. First, a unit mismatch: the trial’s data table is headed “phenotypic resistance to the tetracycline antibiotic

class” while its methods define the *S. aureus* assay as doxycycline, and the federal guideline 122
then describes the doxycycline result as “tetracycline resistance in *S. aureus*”.² Because ef- 123
flux and ribosomal-protection mechanisms differ in whether doxycycline still works (below), 124
an endpoint scored as “the tetracycline class” does not answer the treatment-relevant question. 125
Second, the signal was reported over differing, incompletely documented denominators and 126
then propagated into secondary reporting under further contrasts (Table 1): the resistant nu- 127
merator changes from 5 to 16 to 28 and the denominator basis shifts from colonized carriers to 128
all-swabbed, while the percentage travels more reliably than the denominator, contrast and ter- 129
minology needed to interpret it. The individual-level data that could reconcile these first-party 130
reports are not publicly available.¹ Even the intensively surveilled gonococcus was phenotyped 131
in fewer than one diagnosis in five (resistance testing unavailable for 83%, 212/256).¹⁴ 132

Across trials the phenotype was measured on incompatible terms: DoxyPEP measured doxy- 133
cycline resistance by E-test over two denominators; DuDHS measured it by disc diffusion in 134
single-digit carriers; DOXYVAC measured only MRSA carriage—a methicillin, not a tetracy- 135
cline, quantity, and one that itself rose within the doxy-PEP arm (1.8% to 9.9%) while being 136
reported as a between-arms null.¹⁵ A pooled estimate is not defensible, and the heterogeneity is 137
itself a finding. The trial resolved the doxycycline phenotype but not its molecular basis—efflux 138
(*tet(K)*) versus ribosomal protection (*tet(M)*)—which a single breakpoint cannot separate.^{3,8} 139

The estimand shift did not propagate downstream: no guidance document in the corpus 140
cites the final time-to-event analysis or any hazard ratio, and the two documents revised after 141
its online publication in March 2025—WHO (2026) and San Francisco (May 2026)—retained a 142
cross-sectional or generic frame. Non-citation is telling here because this corpus demonstrably 143
transmits numbers when it has them: CDC carried the cross-sectional 5%→13% figures and 144
WHO cites Soge’s colonization estimate.¹³ 145

Stream B: *S. aureus* monitoring requirements in governmental guidance

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The governmental guidance shows a consistent within-document asymmetry: the efficacy ques- 147
tion received a graded, systematic evidence review, while the AMR-harm question—explicitly 148
including resistant-pathogen development—received an ungraded review in the same docu- 149

ment. Of the six documented-outbreak jurisdictions with governmental doxy-PEP guidance (San Francisco, Chicago, New York City, Boston/Massachusetts, Los Angeles County and San Diego), 0 require or suggest *S. aureus* monitoring. The decline is not an artifact of copying: four jurisdictions are independently authored, one (Los Angeles County) is adapted from San Francisco and one (Massachusetts) from an external template. Nor is it for want of naming the organism—three of the six name *S. aureus* specifically, two invoke only a generic “microbiome” risk and one is silent; even where named, none measure it. Across the full coded corpus (10 documents: CDC, WHO and eight US jurisdictions) the result is identical: 0 instrument the bystander. Raw agreement between the two coding passes was 92%; the single field with meaningful disagreement ($\kappa = 0.58$, context only) is reported unrounded. Providers are nonetheless required to counsel patients about resistance in commensal organisms—a harm no system measures.

Stream C: surveillance capability and exposure–phenotype linkage

Following the signal requires one system to carry, at a common population denominator, both who is exposed to doxy-PEP and what tetracycline phenotype *S. aureus* carries. Across the closed universe of twelve established US systems, none does (Figure 1). Two systems capture exposure at a population denominator (AIDSVu and CDC AtlasPlus, as a pre-exposure prophylaxis proxy, not doxy-PEP itself),¹⁸ and zero capture *S. aureus* tetracycline resistance at a population denominator: the join is vacuous because one half does not exist.

The absence is a codified seam. Wherever *S. aureus* is surveilled at a population denominator the reported axis is methicillin: the one population-based platform, Active Bacterial Core surveillance/Emerging Infections Program (ABCs/EIP), reports invasive MRSA; the National Healthcare Safety Network reports MRSA events; and the only routine source of *S. aureus* tetracycline susceptibility, the facility cumulative antibiogram, is non-standardized and carries no population denominator, because CLSI M39 mandates the MRSA/MSSA split and makes tetracycline a selectively suppressible supplemental agent. The National Antimicrobial Resistance Monitoring System excludes *S. aureus* altogether; tetracycline-resistant *S. aureus* is nationally notifiable nowhere. Exposure exists at county/metro resolution and phenotype only

at denominator-free facility level, so no geography carries both. A system could be built—most
directly by adding *S. aureus* tetracycline susceptibility to ABCs/EIP, which already holds the
denominator—but none is deployed.

Nor could an ordinary population system recover the missing side by collecting more iso-
lates. Indexing the exposed population on the combined PrEP and people-living-with-HIV male
count—the primary specification, because the trial enrolled both (Materials and methods)—the
realistic-cell within-exposed effect required to move a population rate runs to $RR \approx 6\text{--}28$ across
the design-effect range (Figure 2); the more restrictive PrEP-only indexing gives $RR \approx 14\text{--}64$,
which overstates infeasibility. Either way it is far above any plausible effect. Detection appears
in a single cell of the entire grid, where every generous assumption is maxed at once (55%
uptake, $5\times$ enrichment, 13% baseline, 100 000 isolates/geography-year and optimistic panel
independence): there RR_{needed} falls to 1.03, and under single-comparison inference even the
most favourable cell stays above the yardstick (1.79). The verdict thus turns on parameters no
deployed system reports—exposure density, isolate enrichment and dosing frequency—which
is the measurement gap in miniature: the metropolitan best cell needs $1.8\times$ the combined male
exposure density of the densest US geography, and realistic cells hundreds of times it. The only
configuration that restores sensitivity is to sample *S. aureus* directly from doxy-PEP-exposed
persons, which converts surveillance into a sentinel cohort.

Alt text: A matrix of twelve US surveillance systems against exposure and phenotype capabil-
ity; shading shows that exposure is captured only as a PrEP proxy and the *S. aureus* tetracycline
phenotype is captured at a population denominator nowhere, so no column links both rows.

Alt text: A logarithmic relative-risk axis with a shaded benchmark band at 1.42 to 2.25; the
trial hazard ratio 3.89 sits above the band, surveillance detectability thresholds of about 14 to
64 sit far to the right, and guidance has no threshold.

Literature coverage of the bystander organism

A defined PubMed search (2015–2026) identified 361 doxy-PEP papers, of which 13 (3.6%)
mention a bystander staphylococcal organism and only 2 measure tetracycline-resistant *S. au-*
reus against a doxy-PEP exposure contrast. The bystander is rarely mentioned and, when men-

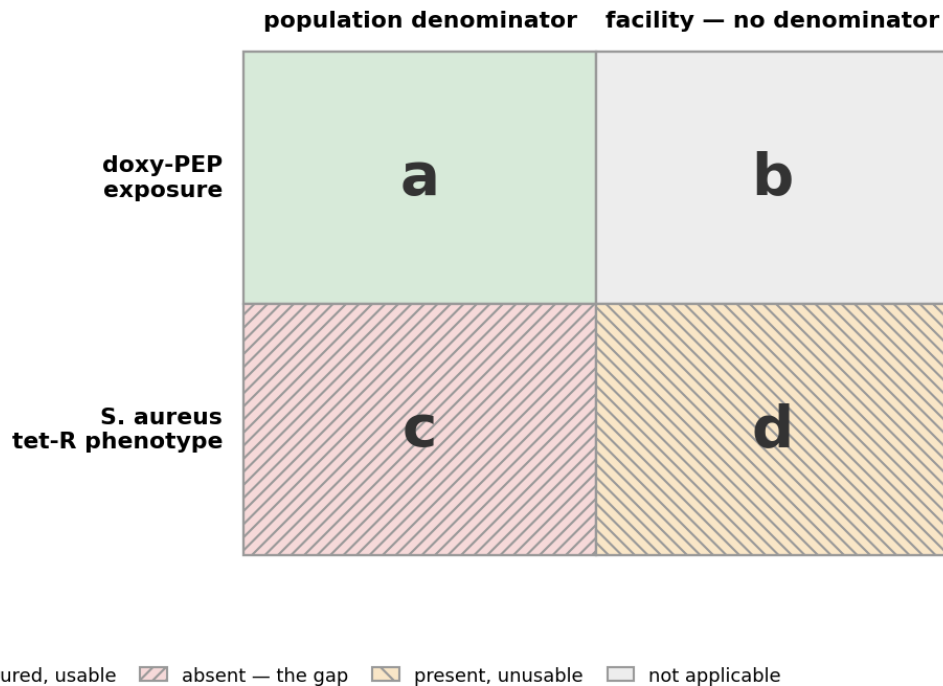


Figure 1: No deployed US system links doxy-PEP exposure to *S. aureus* tetracycline resistance at a common population denominator. Linking the two requires one column usable in both the exposure and phenotype rows; none is. Exposure is captured only as a PrEP proxy (AIDSVu, CDC AtlasPlus); the *S. aureus* tetracycline phenotype is captured at a population denominator nowhere, because the one population-based platform reports the methicillin (MRSA) subset. Of 12 systems, 0 join the two.

tioned, rarely measured against exposure—the research attention concentrated where the exist-
 ing instruments already point. Table 2 summarizes the three streams against the yardstick.

Discussion

The pivotal trial has now resolved the bystander question it was long thought unable to reach:
 a significant increase in incident doxycycline-resistant *S. aureus* under randomized exposure,
 with no matching effect on carriage, and a call from its authors for surveillance of exactly
 this.¹¹ The upstream instrument generated the information. Across the two systems that would
 have to receive it—guidance and deployed surveillance—the answer is that they cannot: guid-
 ance instruments nothing (0/6; 0/10) and no deployed system joins exposure to the *S. aureus*
 tetracycline phenotype at a common denominator (0/12).

The sharper reading is that the trial’s own measurement system evolved to meet its internal

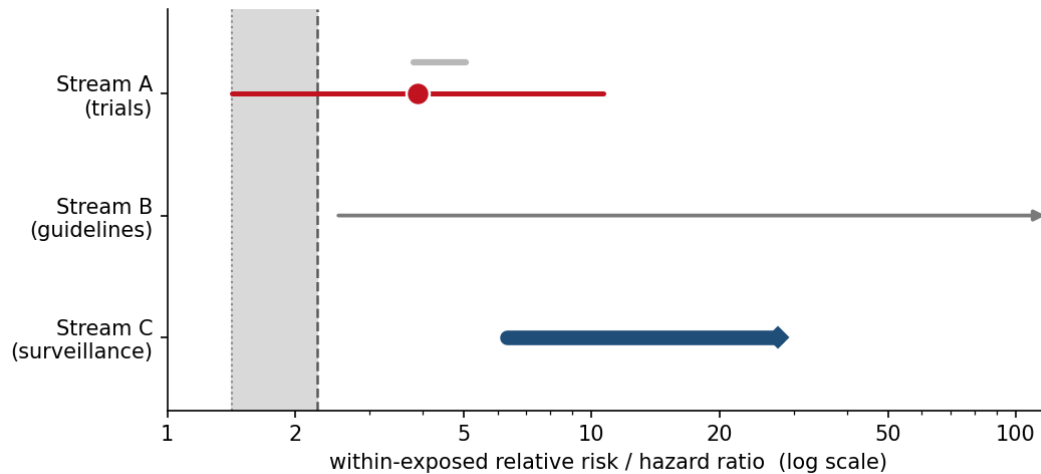


Figure 2: One axis—within-exposed relative risk (log scale)—carries all three streams against the plausible benchmark band (RR 1.42–2.25). The trial’s resolved effect (HR 3.89) clears the band; guidance requires no *S. aureus* measurement so no detection threshold exists; and deployed surveillance would need RR $\approx$ 6–28 to move a population rate (combined PrEP+PLWH exposed denominator; the more restrictive PrEP-only indexing gives 14–64). The signal is above the line while the systems built to follow it are not.

question while the implementation system did not. The 2023 interim was presented as reas- 217
suring; the 2024 guideline appended a call for monitoring; the 2025 final analysis asked the 218
estimand the others had not and returned HR 3.89. Three years, one underlying phenotype, 219
an evolving estimand¹⁹—and, at the end, a demonstrated signal handed to systems built to 220
measure something else. 221

Tetracycline resistance mechanisms and the phenotype unit 222

Staphylococcal surveillance is standardized around methicillin at every tier, yet the treatment- 223
relevant axis here is tetracycline susceptibility, used empirically before methicillin status is 224
known. The two are not interchangeable, and neither is the molecular basis of tetracycline 225
resistance: *tet(K)* efflux leaves doxycycline largely usable, whereas *tet(M)* ribosomal protec- 226
tion can compromise it, and the trial resolved incident doxycycline resistance without resolving 227
which.^{3,8} The distinction is not academic. In a national surveillance population of 823 invasive 228
USA300 isolates, 72 (9%) were tetracycline-resistant, yet 69 were doxycycline-susceptible and 229
carried *tet(K)* while only 3 were doxycycline-resistant and carried *tet(M)*²⁰: a single tetracy- 230
cline breakpoint reports one number for two clinically opposite situations. The mechanisms 231

also differ in mobility—*tet*(M) is characteristically carried on Tn916-like conjugative trans-
posons whose host range includes *S. aureus* and which transfer across strain boundaries by
cell-to-cell contact,²¹ while the *tet*(K) efflux determinant is typically plasmid-borne^{6,20}—so a
clonally indexed system can track a lineage faithfully and still miss a determinant moving be-
tween lineages. An exposure cohort shows the same blindness directly: among 168 military
personnel, doxycycline antimalarial prophylaxis was not associated with overall tetracycline-
class resistance in *S. aureus*, yet *tet*(M) genes were significantly more common in the exposed
($p=0.031$)²²—the aggregate phenotype null while the determinant distribution moved under
exposure (a military, antimalarial-dosed cohort, not doxy-PEP). A surveillance variable can re-
main valid for the question it was designed to answer while being insufficient for the one now
imposed on it. We do not forecast which mechanism is spreading; the point is that the inherited
variable cannot say.

Instrument requirements for population monitoring

Following the signal to the population requires an instrument the deployed architecture does not
provide: sentinel sampling of *S. aureus* from doxy-PEP-exposed individuals, which raises the
exposed fraction by construction and fixes the denominator; sampling that includes the oropharyngeal and peri-anal sites, since nares-only screening under-detects colonization by about 20%
in this population²³; resistance reported per carrier and by mechanism (*tet*(M) versus *tet*(K))
rather than as a pooled tetracycline-class rate; dosing frequency rather than prescription status;
a feed carrying *S. aureus* tetracycline susceptibility at a population denominator, most directly
by adding it to the platform that already tracks invasive MRSA and joining it to exposure on
geography; and access to individual-level trial data. None is exotic; none is in place at scale.

These failures share a structure. A guideline or a surveillance system carries in its sampling
frame, cadence, coding, denominator and unit the organism and question it was built for. The
doxy-PEP apparatus was built to measure gonococcal efficacy; applied to a bystander com-
mensal it inherits a denominator the intervention itself moves, a phenotype label spanning two
mechanisms, and a surveillance architecture standardized around methicillin while the tetracy-
cline unit falls orthogonally through it. We call this *measurement inheritance*: a downstream

system retains the dimensions—estimand, denominator, unit, cadence, linkage, measurement responsibility—appropriate to the upstream question but not those the downstream question requires, so the question becomes unanswerable not for want of data but for want of an inherited dimension. It is built from familiar parts—construct validity, estimand mismatch, ascertainment and surveillance bias,²⁴ metric substitution of the Goodhart type²⁵—but concerns how such mismatches couple *across* stages of an evidence chain. The downstream gap is therefore not closable by a better analysis of the trial’s data—that analysis has been done, and it is significant—but is a property of the guidance and surveillance instruments.

Doxy-PEP’s benefit accrues to the recipient; the tetracycline selection cost falls on people outside the prescribing decision—the outpatient with a skin infection and, at the sharp end, dialysis, surgical and oncology patients for whom an oral anti-staphylococcal agent is no convenience. A negative externality that no deployed instrument can price is not thereby small; it is unweighable. The wider distributive argument—that the two sides are observed on unequal terms, a property we term *distributive observability*—is developed in the Supplementary data (S6).²⁶

Limitations

We make no population-level causal claim and no transmission claim: whether the individual-level signal propagates into a population externality is precisely what no deployed system can determine. The trial’s own randomized result is significant and we accept rather than dispute it; our earlier aggregate reconstruction of the interim is retired in its favor. Colonization is not infection, and the colonization–infection link is itself unmeasured here. Much of the evidence for how *S. aureus* clusters and moves through these networks exists only for its MRSA subset, so where our clustering analysis relies on MRSA we mark it as such—an instance of the inherited limitation, not an exception to it. Finally, the trial reaches only a roughly twelve-month horizon; persistence, dose–response and mechanism remain open. Our coding carries two acknowledged asymmetries: the guidance corpus was double-coded by two machine passes that share a model lineage (reported as raw concordance, not independent human adjudication), while the surveillance enumeration was coded by a single pass. Consistent with a falsification

standard set in advance, the inheritance claim fails if a doxy-PEP guideline comes to require *S. aureus* monitoring, or if a deployed system links doxy-PEP exposure to an *S. aureus* tetracycline phenotype at a common population denominator.

Conclusions

The randomized instrument generated a bystander-resistance signal that the guidance and surveillance built around doxy-PEP are not instrumented to follow. The gap is one of measurement architecture, not of missing data, and it is closable—by sentinel surveillance that links doxy-PEP exposure to the *S. aureus* tetracycline phenotype at a common denominator. Until it is built, the population question the trial’s authors posed cannot be run.

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A large language model was used for writing and analysis-engineering assistance beyond copy-editing. The scientific argument, the reading of primary sources, the model specification and all conclusions are the author’s own; the author reviewed all generated content, independently verified every numerical value and quotation against primary sources and the regenerable analysis outputs, and takes full responsibility. No artificial-intelligence tool is credited as an author.

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Transparency declarations

A. C. D. reports former employment at Gilead Sciences (2020–2024) with prior ownership of corporate stock, all divested by 2024; Gilead played no role in this work. A. C. D. is sole owner of Nyx Dynamics, LLC. No other conflicts to declare. All data are public (AIDS Vu; published trials and guidelines) and all analysis code is openly available and archived with a

citable DOI, regenerating every figure and table by a single command; every coded source is
SHA-256-pinned. No restricted-access data were used.

References

- [1] Luetkemeyer AF, Donnell D, Dombrowski JC, et al. Postexposure doxycycline to prevent
bacterial sexually transmitted infections. *N Engl J Med*. 2023;388:1296-306.
- [2] Bachmann LH, Barbee LA, Chan P, Reno H, Workowski KA, Hoover K, et al. CDC
clinical guidelines on the use of doxycycline postexposure prophylaxis for bacterial sex-
ually transmitted infection prevention, United States, 2024. *MMWR Recomm Rep*.
2024;73(2):1-8. Doi:10.15585/mmwr.rr7302a1. Verified quotes: 'other common bacte-
rial pathogens (e.g., *Staphylococcus aureus*)'; increase in tetracycline-resistant *S. aureus*
'from 5% [20 of 428] to 13% [28 of 222]'; 'potential risks ... will need to be monitored
as these guidelines are implemented'.
- [3] Grossman TH. Tetracycline antibiotics and resistance. *Cold Spring Harb Perspect Med*.
2016;6(4):a025387.
- [4] Diep BA, Chambers HF, Graber CJ, Szumowski JD, Miller LG, Han LL, et al. Emer-
gence of multidrug-resistant, community-associated, methicillin-resistant *Staphylococcus*
aureus clone USA300 in men who have sex with men. *Ann Intern Med*. 2008;148(4):249-
57. Doi:10.7326/0003-4819-148-4-200802190-00204. Chambers (2nd author) also au-
thored the IDSA MRSA guideline [liu2011]; senior author Perdreau-Remington. Not in
hashed corpus; verified via PubMed PMID 18283202.
- [5] de Jong N, et al. Community-acquired methicillin-resistant *Staphylococcus aureus* (CA-
MRSA) in men having sex with men (MSM): a systematic review of risk factors, preva-
lence, and transmission. *BMC Infect Dis*. 2025;25:299. Doi:10.1186/s12879-025-10593-
3.
- [6] Han LL, McDougal LK, Gorwitz RJ, Mayer KH, Patel JB, Sennott JM, et al. High fre-
quencies of clindamycin and tetracycline resistance in methicillin-resistant *Staphylococ-*

- cus aureus pulsed-field type USA300 isolates collected at a Boston ambulatory health center. *J Clin Microbiol.* 2007;45(4):1350-2.
- [7] Stevens DL, Bisno AL, Chambers HF, Dellinger EP, Goldstein EJC, Gorbach SL, et al. Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America. *Clin Infect Dis.* 2014;59(2):e10-52. Doi:10.1093/cid/ciu296.
- [8] Liu C, Bayer A, Cosgrove SE, Daum RS, Fridkin SK, Gorwitz RJ, et al. Clinical practice guidelines by the Infectious Diseases Society of America for the treatment of methicillin-resistant *Staphylococcus aureus* infections in adults and children. *Clin Infect Dis.* 2011;52(3):e18-55. Doi:10.1093/cid/ciq146. Verified in corpus: oral CA-MRSA options include a tetracycline (doxycycline or minocycline), A-II; tet(K) confers tetracycline resistance and inducible doxycycline resistance, tet(M) resistance to all agents in the class.
- [9] Donà MG, Zaccarelli M, Giuliani E, et al. Doxycycline postexposure prophylaxis: a cross-sectional survey on knowledge, attitudes and prescribing practices among Italian dermatologists of sexually transmitted infection centers. *Ital J Dermatol Venereol.* 2026;161(2):120-5. Doi:10.23736/S2784-8671.25.08346-X. n=36; MRSA selection and *N. gonorrhoeae* resistance tied as top concerns at 91.7% each.
- [10] Spence M, Fowler C, Absalom S, Roper T, Williams D, Richardson D. Healthcare professionals' beliefs and concerns about the use of doxycycline post-exposure prophylaxis (doxyPEP): a systematic review. *Venereology.* 2026;5(1):9.
- [11] Luetkemeyer AF, Donnell D, Cohen SE, et al. Doxycycline to prevent bacterial sexually transmitted infections in the USA: final results from the DoxyPEP multicentre, open-label, randomised controlled trial and open-label extension. *Lancet Infect Dis.* 2025;25:873-83.
- [12] Armond ACV, Cobey KD, Moher D. Research Integrity definitions and challenges. *Journal of Clinical Epidemiology.* 2024;171:111367.

- [13] Soge OO, Thibault CS, Cannon CA, et al. Potential impact of doxycycline post-exposure prophylaxis on tetracycline resistance in *Neisseria gonorrhoeae* and colonization with tetracycline-resistant *Staphylococcus aureus* and Group A *Streptococcus*. *Clin Infect Dis*. 2025;80:1188-96.
- [14] Luetkemeyer AF, DoxyPEP Study Team. DoxyPEP and antimicrobial resistance in *N. gonorrhoeae*, commensal *Neisseria* and *S. aureus*. Seattle, WA; 2023. Abstract body and data table; no DOI (CROI abstracts are not assigned DOIs). Conference on Retroviruses and Opportunistic Infections (CROI), Oral Abstract OA-3.
- [15] Vanbaelen T, Manoharan-Basil SS, Kenyon C. Studies of postexposure prophylaxis with doxycycline should consider population-level selection for antimicrobial resistance. *Lancet Infect Dis*. 2024;24:e606-7. Doi:10.1016/S1473-3099(24)00502-4. Reanalysis of DOXYVAC MRSA carriage: within-arm trend significant in both arms.
- [16] Mayer KH, Traeger M, Marcus JL. Doxycycline Postexposure Prophylaxis and Sexually Transmitted Infections. *JAMA*. 2023;330(14):1381-2.
- [17] Flores J, Davis AM, Hazra A. Doxycycline Postexposure Prophylaxis to Prevent Bacterial Sexually Transmitted Infection. *JAMA*. 2025;333(3):248-9.
- [18] AIDSVu. State and county PrEP use and PrEP-to-Need Ratio, 2012–2025; 2026. Public data; accessed 2026. Emory University, Rollins School of Public Health. <https://aidsvu.org>.
- [19] Renson A, Montoya L, Goin DE, Díaz I, Ross RK. Pulling Back the Curtain: The Road from Statistical Estimand to Machine-Learning-Based Estimator for Epidemiologists (No Wizard Required). *American Journal of Epidemiology*. 2025;194(12):3566-71.
- [20] McDougal LK, Fosheim GE, Nicholson A, Bulens SN, Limbago BM, Shearer JES, et al. Emergence of resistance among USA300 methicillin-resistant *Staphylococcus aureus* isolates causing invasive disease in the United States. *Antimicrob Agents Chemother*. 2010;54(9):3804-11.

- [21] Roberts AP, Mullany P. Tn916-like genetic elements: a diverse group of modular mobile 389
elements conferring antibiotic resistance. *FEMS Microbiol Rev.* 2011;35(5):856-71. 390
- [22] Mende K, Beckius ML, Zera WC, Yu X, Li P, Tribble DR, et al. Lack of doxycycline 391
antimalarial prophylaxis impact on *Staphylococcus aureus* tetracycline resistance. *Diagn* 392
Microbiol Infect Dis. 2016;86(2):211-20. 393
- [23] Kyaw WM, Lee LK, Siong WC, Ping ACL, Ang B, Leo YS. Prevalence of and risk 394
factors for MRSA colonization in HIV-positive outpatients in Singapore. *AIDS Res Ther.* 395
2012;9:33. 396
- [24] Tancredi S, Cullati S, Chiolerio A. Surveillance bias in the assessment of the size of 397
COVID-19 epidemic waves: a case study. *Public Health.* 2024;234:98-104. 398
- [25] Treem JW, Barley WC, Weber MS, Barbour JB. Signaling and Meaning in Organiza- 399
tional Analytics: Coping with Goodhart's Law in an Era of Digitization and Datafication. 400
Journal of Computer-Mediated Communication. 2023;28(4):zmad023. 401
- [26] Montoya LM, Geng EH, Valancius M, Kosorok MR, Petersen ML. Effects Among the 402
Affected. *Statistics in Medicine.* 2025;44(28-30):e70353. 403

Table 1: Propagation of the DoxyPEP *S. aureus* resistance signal across reporting venues. Rows are not assumed to be identical estimands; differences in contrast, conditioning set, denominator and terminology are the object of comparison.

Venue	Reported result	Frame / conditioning set
NEJM appendix ¹	5/31 = 16%	Month-12 doxy-PEP arm; resistant among carriers
CROI abstract ¹⁴	16/137 = 12%	Doxy-PEP arm; all-swabbed denominator
CDC guideline ²	28/222 = 13%; summarized 5%→13%	Doxy-PEP arm; guideline frame
JAMA Insights ¹⁶	5% versus 4%	Doxy-PEP versus control; “infected with”
JAMA Synopsis ¹⁷	5%→16%	Within arm, among <i>S. aureus</i> -isolated

Table 2: Headline audit of the doxy-PEP evidence base against one detectability yardstick. Each cell is licensed by the Results above; no composite score is implied.

Stream	Question	Result
A. Trials	Was the bystander signal resolved?	Yes—HR 3.89 (1.42–10.68), incident doxy-R <i>S. aureus</i>
B. Guidance	Is <i>S. aureus</i> monitoring required?	No—0/6 outbreak-matched; 0/10 coded
C. Surveillance	Is exposure linked to phenotype?	No—0/12 systems; $RR_{\text{needed}} \approx 6\text{--}28$ to detect